CUSTOMER SHOPPING BEHAVIOR ANALYSIS

REPORT

Data Analysis Project

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Tools Used: Python, PostgreSQL – Admin 4, Microsoft Power Bi, Microsoft Excel

1. Abstract

This project analyzes a retail dataset of 3,900 customer transactions to understand shopping behavior. The data was cleaned and feature-engineered in Python, loaded into PostgreSQL, and analyzed with 10 business queries. The findings were then presented in an interactive Power BI dashboard. Total revenue was $233,081, with an average purchase of $59.76 and an average review rating of 3.75. Clothing is the top revenue category. 80% of customers are “Loyal” repeat buyers, and subscribers spend about the same as non-subscribers. The report closes with recommendations on subscriptions, discounts and marketing.

2. Introduction

Retailers need to know who their customers are, what they buy, and what drives spending in order to plan inventory, pricing and promotions. This project applies the data analytics workflow of cleaning, storing, querying and visualizing to a customer shopping dataset.

3. Objectives

- Clean and prepare the raw dataset for analysis.
- Create new features (age group, purchase frequency in days).
- Answer 10 business questions using SQL.
- Build an interactive Power BI dashboard of key metrics.
- Summarize insights and recommendations.

4. Dataset Description

- **Records:** 3,900 transactions
- **Columns:** 18 (5 numeric, 13 categorical)
- **Format:** CSV

Column	Description
Customer ID	Unique customer identifier
Age	Customer age (18–70)
Gender	Male / Female
Item Purchased	25 unique products
Category	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	Transaction value ($20–$100)
Location	50 U.S. states
Size, Color	Product attributes
Season	Season of purchase
Review Rating	Customer rating
Subscription Status	Yes / No
Shipping Type	6 shipping options
Discount Applied	Yes / No
Promo Code Used	Yes / No
Previous Purchases	Count of past purchases
Payment Method	6 payment methods
Frequency of Purchases	How often the customer buys

Summary statistics: mean purchase amount $59.76 (median $60, std. dev. $23.69), mean age 44.07, and 68% male customers (2,652 of 3,900).

5. Methodology

5.1 Data Cleaning and Preparation (Python)

1. **Loaded the data** with Pandas and inspected it using `head()`, `info()` and `describe()`.
2. **Handled missing values.** Only Review Rating had nulls (37 of 3,900). They were filled with the median rating of each product category, so no rows were lost.
3. **Standardized column names** to lowercase with underscores (e.g., `Purchase Amount (USD)` became `purchase_amount`).

4. **Created age_group** using quartile binning (pd.qcut) into four groups: Young Adult, Adult, Middle-age and Senior.
5. **Created purchase_frequency_days** by mapping frequency labels to days (Weekly = 7, Fortnightly/Bi-Weekly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, Annually = 365).
6. **Removed a redundant column.** promo_code_used was identical to discount_applied in every row, so it was dropped.

5.2 Database Integration (PostgreSQL)

The cleaned DataFrame was loaded into a PostgreSQL database (customer_behavior, table customer) using SQLAlchemy and psycopg2. All business questions were then answered with SQL.

5.3 Visualization (Power BI)

An interactive dashboard was built with KPI cards, charts and slicers for Subscription Status, Gender, Category and Shipping Type.

6. Analysis and Findings

Q1. Revenue by gender

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Male customers generate about 68% of revenue, which is in line with their 68% share of customers. Average spend per purchase is nearly equal (Female $60.25, Male $59.54).

Q2. Discount users who spent above average

839 of the 1,677 discounted purchases (50%) were at or above the average purchase amount ($59.76). This shows discounts do not stop customers from making high-value purchases.

Q3. Top 5 products by average review rating

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Q4. Average spend by shipping type

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Express shipping customers spend about $2 more per order.

Q5. Subscribers vs. non-subscribers

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Subscribers are 27% of customers and contribute about 27% of revenue. Their average spend is essentially the same as non-subscribers' ($59.49 vs. $59.87).

Q6. Top 5 products by discount rate

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

Q7. Customer segmentation

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Which means Loyal Shares are 79.9% and Returning are 18.0% and New ones are 2.1%

Q8. Top products per category (by orders)

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Category	**#1**	**#2**	**#3**
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)

Q9. Repeat buyers (more than 5 previous purchases) and subscription

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Of 3,476 repeat buyers, 958 (27.6%) subscribe and 2,518 do not. This is almost the same as the overall subscription rate of 27%, so repeat buyers are **not** noticeably more likely to subscribe.

Q10. Revenue by age group

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-age	59197
3	Adult	55978
4	Senior	55763

Additional findings from the dataset

- **Revenue by category:** Clothing $104,264 (44.7%), Accessories $74,200 (31.8%), Footwear $36,093 (15.5%) and Outerwear $18,524 (7.9%).
- **Seasonality:** Revenue is evenly spread, with Fall highest ($60,018) and Summer lowest ($55,777).
- **Discounts:** 43% of purchases used a discount. Average spend with a discount is $59.28, versus $60.13 without.
- **Top states by revenue:** Montana, Illinois, California, Idaho and Nevada.

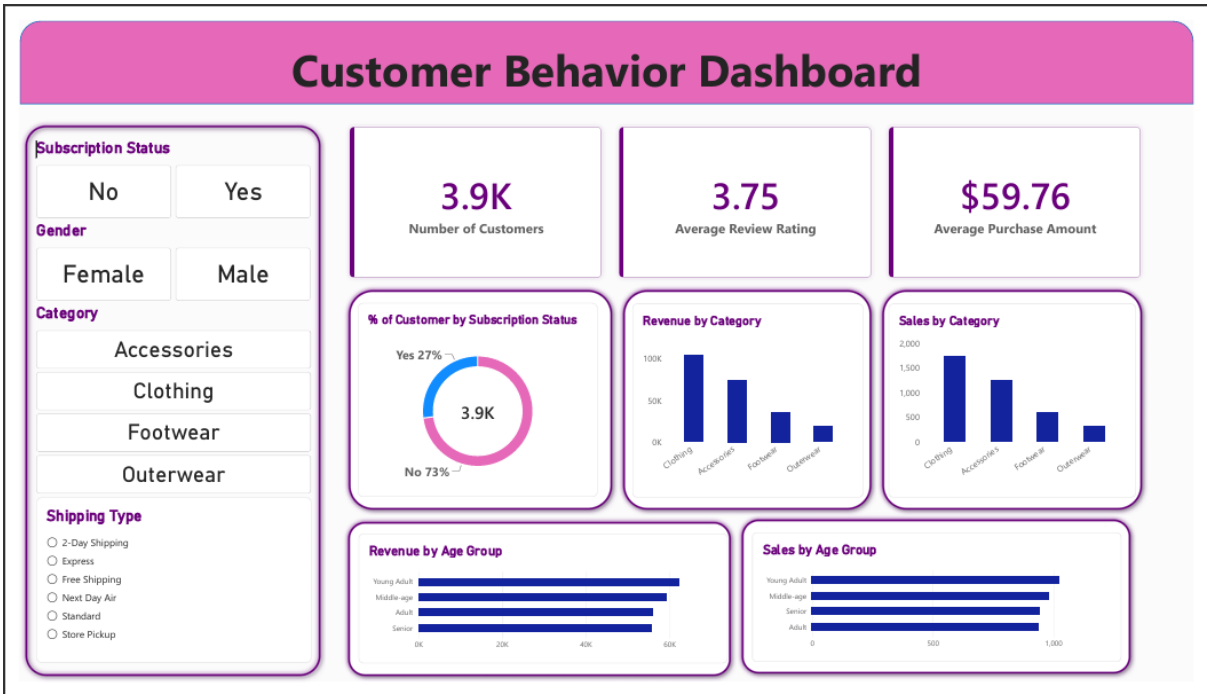
7. Power BI Dashboard

Key metrics (KPI cards):

- Number of customers: 3.9K
- Average review rating: 3.75
- Average purchase amount: $59.76

Visuals: subscription share donut (Yes 27%, No 73%), revenue by category, sales by category, revenue by age group, and sales by age group.

Interactivity: slicers for Subscription Status, Gender, Category and Shipping Type.



8. Insights and Recommendations

1. **Focus on Clothing and Accessories.** They bring in over 76% of revenue, so prioritize inventory and marketing for them.
2. **Rethink the subscription program.** Subscribers do not spend more, so add perks such as exclusive products or free shipping to make it worthwhile.
3. **Use discounts selectively.** Discounts do not raise average spend and about half of all discounted buyers already spend above average, so target them at slow-moving items.
4. **Grow the "New" segment.** Only 2% of customers are new, so acquisition campaigns are needed.
5. **Target Young Adults.** They are the top revenue group, so use social media and trend-based campaigns for them.
6. **Promote Express shipping.** Express users spend more per order.
7. **Improve female customer acquisition.** Women are only 32% of customers but spend as much per order as men.
8. **Reward loyal customers.** With 80% of customers in the Loyal segment, a loyalty-points program can protect this base.

9. Conclusion

The project covered the full analytics pipeline of cleaning, storing, querying and visualizing. The analysis shows a loyal, evenly spending customer base. Revenue depends on category mix and customer volume rather than on differences in spend per order between groups. Subscriptions and discounts do not increase spending in this data, so the retailer should rethink how they are used.

10. Limitations and Future Scope

Limitations:

- The dataset has no date column, so true time-series analysis was not possible.
- Review ratings were imputed for 37 records.

Future scope:

- Customer clustering (K-Means) for segmentation.
- Predictive models for subscription or high-spend customers.
- Real-time dashboard connected directly to PostgreSQL.